

FOUNDATIONAL MATHEMATICS NOTE

Toward a Two-Class Foundation of Mathematical Objects

A formal presentation of the proposed distinction between surfaces and laws as mutually exclusive primitive classes.

Primitive Symbols

- $\mathcal{S}$ denotes the class of surfaces.
- $\mathcal{L}$ denotes the class of laws.
- $\mathcal{U}$ denotes the universe of mathematical objects.

Axioms

Axiom 1 — Completeness

Every primitive mathematical object belongs to exactly one class.

$$\forall x \in \mathcal{U}, \quad x \in \mathcal{S} \vee x \in \mathcal{L}.$$

Axiom 2 — Mutual Exclusivity

No primitive object is simultaneously a surface and a law.

$$\forall x \in \mathcal{U}, \quad \neg (x \in \mathcal{S} \wedge x \in \mathcal{L}).$$

Corollary 2.1

The two classes form a disjoint partition of the universe.

$$\mathcal{U} = \mathcal{S} \sqcup \mathcal{L}.$$

Axiom 3 — Transformation

Every admissible transformation between surfaces is induced by at least one law.

$$S_1, S_2 \in \mathcal{S}, \quad S_1 \rightarrow S_2 \implies \exists L \in \mathcal{L}.$$

Axiom 4 — No Self-Evolution

No surface evolves independently of a law.

$$S_1 \rightarrow S_2 \implies \exists L \in \mathcal{L} \text{ such that } L(S_1) = S_2.$$

Axiom 5 — Irreducibility

No law is identical to any surface.

$$\forall L \in \mathcal{L}, \forall S \in \mathcal{S}, \quad L \neq S.$$

Axiom 6 — Representation

A surface may encode a law without becoming identical to that law.

$$R(L) \in \mathcal{S}, \quad L \in \mathcal{L}, \quad R(L) \neq L.$$

Definitions

Definition 1 — Surface

A *surface* is an object whose identity is completely determined by its intrinsic state. A surface possesses no primitive notion of transformation.

Definition 2 — Law

A *law* is an object whose identity is completely determined by the admissible transformations it defines between surfaces. A law is not itself a surface.

Foundational Principle

Principle 1

The ontology of mathematics consists of two primitive classes:

1. state-bearing objects,
2. transformation-bearing objects.

No third primitive class is assumed.

Conjectures

Conjecture 1

Every mathematical theory admits a canonical decomposition

$$(\mathcal{S}, \mathcal{L}).$$

Conjecture 2

Every physical theory admits a unique decomposition into a family of surfaces and a family of laws.

Conjecture 3

No faithful isomorphism exists between the class of surfaces and the class of laws.

$$\mathcal{S} \not\cong \mathcal{L}.$$

Conjecture 4

Every computable process can be expressed as repeated application of elements of $\mathcal{L}$ acting upon elements of $\mathcal{S}$.

Conjecture 5

The partition

$$\mathcal{U} = \mathcal{S} \sqcup \mathcal{L}$$

is more fundamental than distinctions such as state versus dynamics, object versus morphism, data versus algorithm, manifold versus vector field, and configuration versus evolution.

Formalization note. The framework becomes substantially stronger once the meanings of “primitive,” “intrinsic state,” “admissible transformation,” “representation,” and “faithful isomorphism” are specified inside a chosen

foundational system, such as category theory, dependent type theory, or a two-sorted logical language.

Standalone HTML document rendered with MathJax 3.